Unit 4 · Outcome 1 · Climate Change

# Greenhouse Gas Warming Potential

KK06 VCE Environmental Science · Units 3 & 4 · Exam study summary

Digital Vector EnviroSci · single-topic mastery summary with worked good-vs-bad answers, exam traps, command words and practice.

## A tonne of methane is not a tonne of carbon dioxide

01 The big idea A tonne of methane is not a tonne of carbon dioxide At the Cape Wirreanda Climate Observatory — our fictional monitoring station on Victoria's far south coast, facing the Southern Ocean — a flux tower samples the air drifting in from the land behind it. On any given day it picks up carbon dioxide from a natural-gas peaking plant, methane from the dairy paddocks, nitrous oxide rising off fertilised soil, and a whisper of sulfur hexafluoride leaking from the electricity switchyard. The observatory has a problem familiar to every climate sci

## Defining Global Warming Potential

02 What the number actually means Defining Global Warming Potential Global Warming Potential (GWP) unitless index · reference gas $CO_2$ = 1 · stated over a chosen time horizon (usually 100 years) A measure of how much heat (infrared radiation) one kilogram of a greenhouse gas traps in the atmosphere over its lifetime , relative to one kilogram of carbon dioxide , over a set period of time (the time horizon). Three things are doing the heavy lifting in that definition, and each one is worth marks: $CO_2$ = 1 Carbon dioxide is the benchmark . Every other gas is

## What makes a gas's warming potential big or small

03 The two ingredients What makes a gas's warming potential big or small The VCAA wording — “the infrared radiation a gas absorbs over its lifetime ” — quietly packs in the two factors that set GWP. A potent gas is one that is good at catching heat and sticks around to keep doing it . ① Radiative efficiency — how strongly it absorbs infrared Earth radiates heat upward as infrared (longwave) radiation . Some molecules are shaped so they absorb this radiation very strongly — they vibrate and re-radiate heat back, with each molecule acting like a tiny, very

## Why methane's number changes — the sprinter and the marathon runner

04 The time-horizon twist Why methane's number changes — the sprinter and the marathon runner ◈ Analogy — sprinter vs marathon runner Methane is a sprinter: it traps heat intensely, then it's gone in ~12 years. Carbon dioxide is a marathon runner: modest per kilogram, but it just keeps going for centuries. Over a 20-year race, the sprinter's burst dominates — methane's GWP is around 80 . Stretch the race to 100 years and the sprinter finished long ago while the marathoner is still running, so methane's averaged-out number falls to about 28 . This is why

## Warming potentials of the gases you'll meet

05 The numbers worth knowing Warming potentials of the gases you'll meet You will not be asked to memorise a long table, but you should know the order of magnitude and the pattern : $CO_2$ low but long-lived; methane in the tens; nitrous oxide in the hundreds; the fluorinated industrial gases in the thousands to tens of thousands. These are 100-year values; Australia reports using the IPCC Fifth Assessment Report figures. Greenhouse gas GWP (100 yr) Lifetime Main local source at Cape Wirreanda Carbon dioxide $CO_2$ 1 centuries+ gas peaking plant, vehicles Meth

| Greenhouse gas | GWP (100 yr) | Lifetime | Main local source at Cape Wirreanda |
|---|---|---|---|
| Carbon dioxide $CO_2$ | 1 | centuries+ | gas peaking plant, vehicles |
| Methane $CH_4$ | ~28 | ~12 yr | dairy cattle, wetland, gas leaks |
| Nitrous oxide $N_2O$ | ~265 | ~115 yr | nitrogen fertiliser on soil |
| HFC-134a refrigerant | ~1,300 | ~14 yr | cold-store & car air-con leaks |
| Sulfur hexafluoride $SF_6$ | ~23,500 | ~3,200 yr | high-voltage switchgear |

## Carbon dioxide equivalent ($CO_2$-e)

06 The common currency Carbon dioxide equivalent ($CO_2$-e) GWP lets us convert any gas into the amount of $CO_2$ that would cause the same warming. That converted figure is carbon dioxide equivalent ($CO_2$-e) , and the formula is the simplest in the whole course: ∑ The formula examiners want $CO_2$-e (tonnes) = mass of gas (tonnes) × GWP The Cape Wirreanda team needs one regional total. Set each source's annual emissions and watch the raw tonnes convert into a fair, comparable $CO_2$-e total. Notice how a small mass of a high-GWP gas can outweigh a large mass of a lo

## Potency is not the same as contribution

07 The distinction that separates A's from C's Potency is not the same as contribution Here is the resolution to the $CO_2$ paradox, and the single most-tested idea in this dot point. A gas's warming potential (GWP) is its potency per kilogram . Its actual contribution to warming depends on potency multiplied by how much is emitted . $SF_6$ is fantastically potent but humans release only a trickle of it, so its total contribution is tiny. $CO_2$ is feeble per kilogram, but we pour out tens of billions of tonnes a year, so it dominates real-world warming. Toggle t

## Ways to keep it straight in the exam

08 Three analogies that stick Ways to keep it straight in the exam ◈ Currency exchange GWP is an exchange rate and $CO_2$-e is the common currency . Just as you convert yen, euros and dollars into one currency to add up a travel budget, you convert $CH_4$, $N_2O$ and $SF_6$ into $CO_2$-e to add up an emissions budget. The "rate" for methane is 28-to-1. ◈ Chilli vs rice Potency vs quantity. One chilli ($SF_6$) is far hotter than one grain of rice ($CO_2$). But eat a whole kilogram of rice and a single chilli, and the rice delivers more total heat to your mouth. Per-unit stren

## Which factor is driving this gas's GWP?

09 Check yourself Which factor is driving this gas's GWP? Drag each statement to the factor it describes. Every greenhouse gas's GWP comes from some mix of these two — sorting them sharpens the exam reflex of naming the cause , not just the number. Mainly strong infrared absorption Mainly long atmospheric lifetime s

## What the verb is really asking for

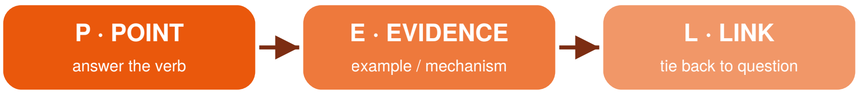
**◈ COMMAND WORDS**

10 Decode the command word What the verb is really asking for Most lost marks come from answering the wrong question. Tap a command word to see exactly what an examiner expects when it appears on a warming-potential item. Tap a command word above…

## The P·E·L method

**✎ P·E·L WRITING**

11 Structure a written answer The P·E·L method For "explain", "compare" and "justify" items, a tidy structure wins marks the content alone won't. P·E·L = Point (state the idea), Example/Evidence (a number, gas or Cape Wirreanda detail), Link (tie back to the question). Point Make the claim in one clean sentence using the key term. Example / Evidence Anchor it with a real value or named gas — GWP figures, lifetimes, a tonne calculation. Link Answer the actual command word — the comparison, the "why", the implication. Worked build — “Explain why methane ha




The P·E·L answer structure — make a Point, support with Evidence, then Link to the question.

## Six ways students throw away marks

12 Named exam traps Six ways students throw away marks ⚠ Trap 1 — "GWP = how much is emitted" GWP is potency per kilogram , independent of how much is released. Quantity is a separate factor. ⚠ Trap 2 — forgetting the benchmark Always anchor to $CO_2$ = 1 . "Methane's GWP is 28" only means anything relative to $CO_2$. ⚠ Trap 3 — ignoring the time horizon A GWP number is meaningless without a horizon. Methane is ~80 over 20 years but ~28 over 100. ⚠ Trap 4 — only naming one factor "Explain GWP" wants both infrared absorption and atmospheric lifetime, not just o

## Five graded questions · 24 marks

**★ PRACTICE & SELF-MARK**

13 Practice exam questions Five graded questions · 24 marks Attempt each in your logbook first, then reveal the weak answer, the strong answer and the marking guide. Mark your own attempt out of the total — your score feeds the dock at the bottom of the page.

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Worked good/weak answers illustrate exam technique — always check against current VCAA marking advice.